

Smart Bike Crash Detection & SOS Module

Automatic emergency response for motorcycle accidents





The Problem

Help Arrives Too Late

Most motorcycle accidents go unreported for critical minutes. Emergency services are not alerted automatically – every second of delay reduces survival odds.

Rider is Incapacitated

A rider who is unconscious, disoriented, or trapped cannot call for help. There is no fallback – no system in place to act on their behalf.

Our Solution

A compact, universal module that mounts on **any motorcycle** – no modifications required. The moment a crash is detected, it acts automatically.



Universal Fit

Attaches to any bike without hardware changes



Live GPS Alerts

Sends exact location to emergency contacts, hospitals & police



Fully Automatic

Zero rider action required – system responds instantly

How It Works



Sensors
Detect

MCU Process

Confirm Crash

10s Cancel

SOS Sent

If the rider is conscious, a **10-second window** allows manual cancellation via a single button press. Otherwise, alerts fire automatically to all emergency contacts.

Key Features



Crash Detection

Automatic impact sensing – no rider input needed



GPS Sharing

Real-time coordinates sent to all recipients



GSM SOS Alert

SMS alerts dispatched via GSM network



Manual Cancel

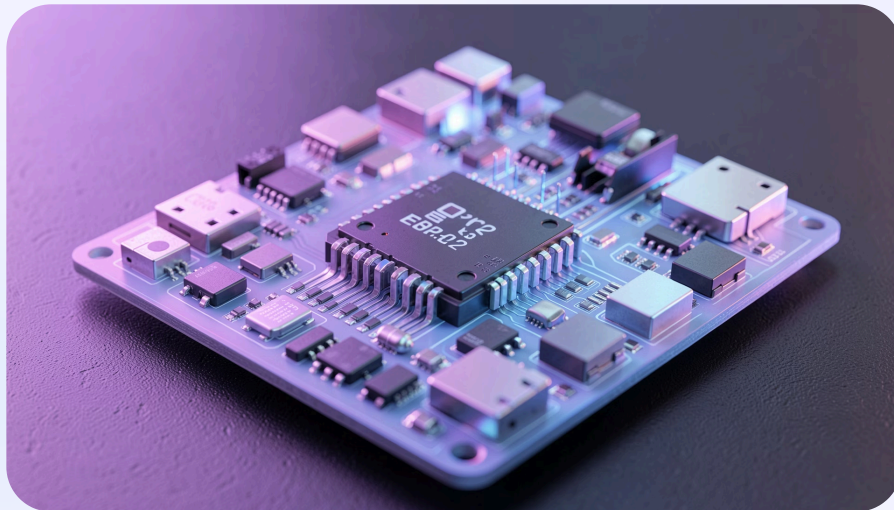
10-second override window for false positives



Any Bike

Fully plug-and-play, model-agnostic installation

MVP Components



1

ESP32

Central microcontroller – WiFi & Bluetooth capable

2

MPU6050

6-axis IMU for impact & tilt detection

3

GPS Module

Live coordinate tracking

4

GSM Module

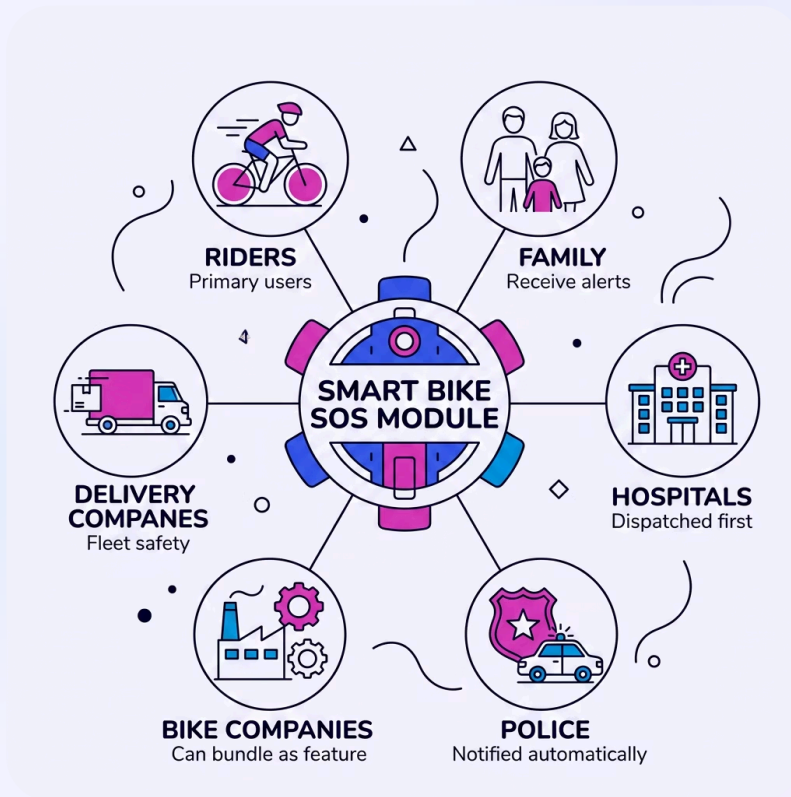
Sends SMS alerts over cellular

5

Battery + Buzzer + Button

Power, audio alert & manual cancel

Who Benefits



From individual riders to large delivery fleets, the module addresses a **wide ecosystem of stakeholders** who all share a common interest: **faster emergency response**.

→ **Riders & Families**
Peace of mind – alerts sent the moment something goes wrong

→ **First Responders**
Hospitals and police receive precise GPS coordinates instantly

→ **Industry Partners**
Bike brands and delivery companies gain a scalable safety layer

Impact & Benefits

⚡ Faster Response

Cuts the critical gap between crash and emergency dispatch

❤️ Saves Lives

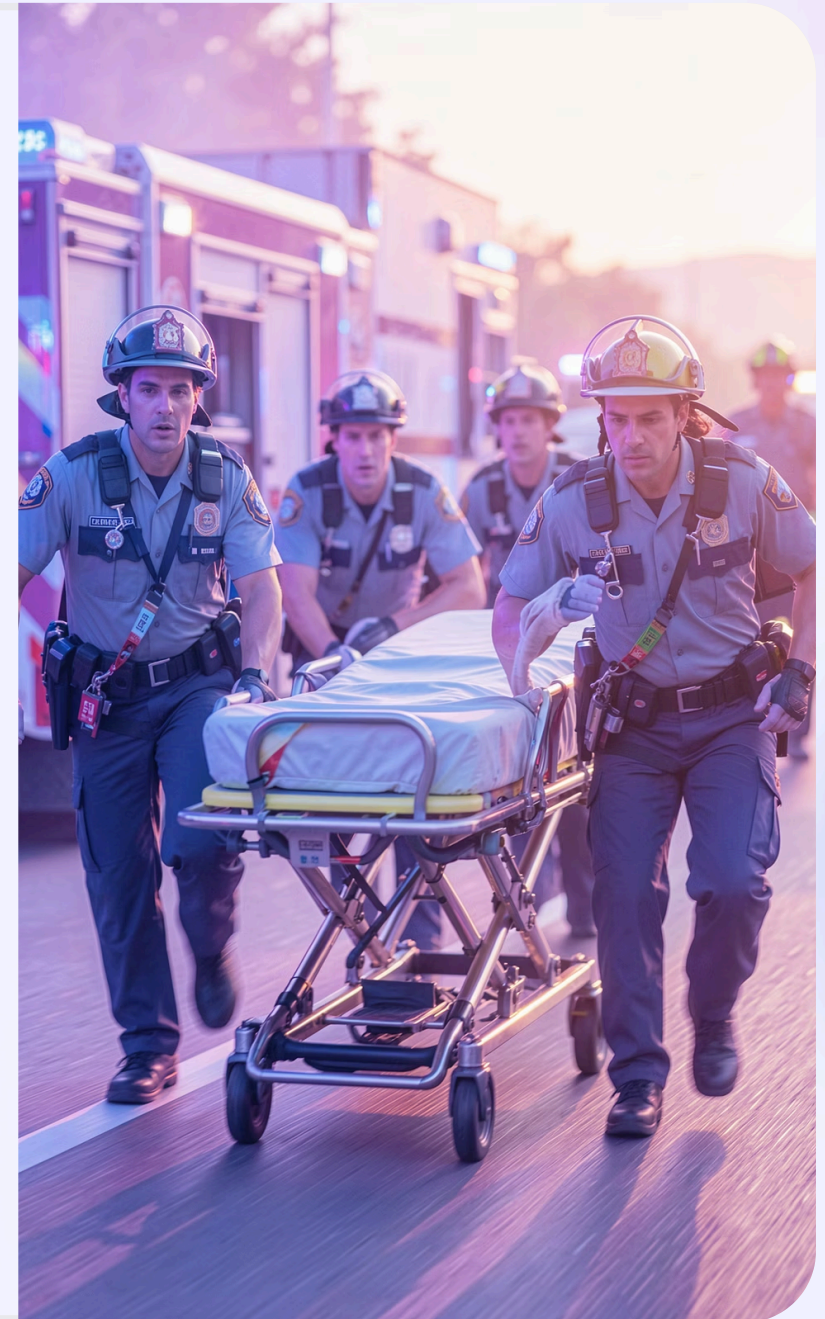
Every minute of response time reduction improves survival odds

💰 Low Cost

BOM under \$30 – accessible globally, scalable by design

🔑 Easy Install

No special tools or expertise – fits any motorcycle in minutes



What's Next



Mobile App Integration

Companion app for alert management, ride history, and contact configuration



AI-Based Crash Detection

Machine learning models trained on real crash data to reduce false positives



Direct Ambulance Integration

API-level connection to emergency dispatch systems for zero-latency response



"Reducing the time
between accident
and help **can save
lives.**"

Smart Bike Crash Detection & SOS Module – built by engineers, for
every rider.